This tutorial assumes a UNIX command line. On Linux or Mac this is built in; on Windows install Cygwin.

Exercise (Setup, MAFFT, HoT, Seaview)

Make a ~/bin directory (for programs) and add it to your PATH

```
% cd                                  // go to your home directory
% pwd                                 // where are we? = print working directory
% ls                                  // look around
% mkdir bin                           // make a subdirectory
% nano .bash_profile                  // edit the startup script
<Add a line "PATH=~/bin:$PATH">
<log out/log back in>
% less .bash_profile                  // examine .bashrc to see if your changes are there
% echo $PATH                          // check that ~/bin is in PATH
```

Download the alignment files we will use:

```
% cd
% mkdir alignment_files
% cd ~/alignment_files
% wget http://www.bali-phy.org/examples.tgz       // Download the files – Linux (On cygwin, first install wget).
% curl -O http://www.bali-phy.org/examples.tgz   // Download the files – Mac
< Alternatively, download the files using your browser >
% tar -zxf examples.tgz               // extract the compressed archive
% ls examples                         // look inside the new directory that was created
% less examples/5S-rRNA/5d.fasta      // examine the downloaded files – type q to quit
% alignment-info examples/5S-rRNA/5d.fasta
```

Mafft and HoT and Seaview

Move the *fasta-flip.pl* script into ~/bin, and align forward and backwards with *mafft*.

```
% cd ~/alignment_files/examples
% chmod +x fasta-flip.pl              // make the file executable
% cp fasta-flip.pl ~/bin              // copy the file to ~/bin
% which fasta-flip.pl                 // check to make sure its in your PATH
% cp 5S-rRNA/25.fasta .               // make a copy of a file
% mafft –auto 25.fasta > 25-mafft.fasta          // try a simple sequence alignment
% fasta-flip.pl 25.fasta | mafft –auto - | fasta-flip.pl > 25-mafft-flipped.fasta // align the sequences when flipped
% seaview 25-mafft.fasta & seaview 25-mafft-flipped.fasta &  // take a look at the two alignments
```

Exercise (muscle, Probcons, SuiteMSA, and consensus)

Make a local copy of a data file

```
% cd ~/alignment_files/examples
% cp HIV/HIVSIV.fasta .                          // Make a copy of the file in the current directory
% alignment-info HIVSIV.fasta                    // Some basic info about the sequences
```

Muscle

Look at the time to perform different numbers of iterations with muscle

```
% time muscle -maxiters 1 < HIVSIV.fasta > HIVSIV-muscle1.fasta      // use a bad guide tree
% time muscle -maxiters 2 < HIVSIV.fasta > HIVSIV-muscle2.fasta      // obtain a better guide tree and realign
% time muscle -maxiters 3 < HIVSIV.fasta > HIVSIV-muscle3.fasta      // also add refinement
% time muscle < HIVSIV.fasta > HIVSIV-muscle.fasta
% muscle -in HIVSIV.fasta -out HIVSIV-muscle.fasta                   // This is the same as the previous command.
```

Compare the first and last alignments
% alignment-cat HIVSIV-muscle1.fasta --reorder-by-alignment=HIVSIV-muscle.fasta > a   // reorder sequences
% mv a HIVSIV-muscle1.fasta
% seaview HIVSIV-muscle1.fasta & seaview HIVSIV-muscle.fasta &

Probcons

Align sequences and reversed sequences with *probcons*
% probcons HIVSIV.fasta > HIVSIV-probcons.fasta
% fasta-flip.pl HIVSIV.fasta > HIVSIV-flipped.fasta
% probcons HIVSIV-flipped.fasta | fasta-flip.pl > HIVSIV-probcons-flipped.fasta

SuiteMSA

Compare alignments using SuiteMSA
% SuiteMSA &
<Choose Pixel Plot, and select HIVSIV-probcons.fasta>
<Click on "Add" to load a new alignment file, and choose HIVSIV-probcons-flipped.fasta>
<Click on "Add" to load a new alignment file, and choose HIVSIV-muscle.fasta>
Q: What is the spatial pattern of the differences?

Consensus

Construct an alignment that only contains homologies present in both mafft and probcons alignments
% echo > blank
% cat blank // It should print a blank line to the screen.

% cat HIVSIV-muscle.fasta blank HIVSIV-probcons.fasta > two-alignments.fasta
% alignment-consensus –strict=0.99 < two-alignments.fasta > consensus.fasta
% seaview consensus.fasta &

Exercise (FSA and Prank)

FSA

Align sequences with *fsa*. This will take some time.
% cd ~/alignment_files/examples
% fsa --fast --gapfactor 0 HIVSIV.fasta > HIVSIV-fsa0.fasta &   // This is similar to probcons – few gaps.
% fsa --fast --gapfactor 0.5 HIVSIV.fasta > HIVSIV-fsa.fasta &   // This is the default. Some gaps.

Look at the alignment lengths for the different *fsa* parameters:
% alignment-info HIVSIV-fsa0.fasta
% alignment-info HIVSIV-fsa.fasta

Prank

Align a (short) 5S data set with *prank*

% cd ~/alignment_files/examples/
% prank -d=ITS/ITS1-trimmed.fasta +F
% cp output.2.fas ITS1-prank1.fasta

% mafft ITS/ITS1-trimmed.fasta > ITS1-mafft.fasta
% muscle < ITS/ITS1-trimmed.fasta > ITS1-muscle.fasta
% SuiteMSA &
<Compare ITS1-prank.fasta, ITS1-mafft.fasta, and ITS1-muscle.fasta using SuiteMSA>